JAVA PROGRAMMING PRACTICES

Prime Palindrome Check

Basic Information

Week	Week 3
Source file	PrimePalindrome.java
Concepts	Variables, operators, loops, conditions, console input/output.

Objective / Aim

To implement prime palindrome check using the concepts listed in the R23 Java Programming Practices syllabus.

System Requirements

JDK (Java Development Kit) and a text editor or Java IDE.

Algorithm / Logic Description

1. Read an integer.
2. Check whether it is prime.
3. Reverse the number and compare it with the original.
4. Display the result.

UML / Program Structure

```
PrimePalindrome
main() demonstrates the algorithm above.
```

Source Code

```
import java.util.Scanner;
public class PrimePalindrome
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int number = sc.nextInt(), original = number;
        boolean prime = number > 1;
        for (int i = 2; i * i <= number; i++)
        {
            if (number % i == 0)
            {
                prime = false;
                break;
            }
        }
        int reverse = 0;
        while (number > 0)
        {
            reverse = reverse * 10 + number % 10;
            number /= 10;
        }
        if (prime && original == reverse) System.out.println(original + " is a prime palindrome");
        else System.out.println(original + " is not a prime palindrome");
        sc.close();
    }
}
```

Compilation and Execution

```
javac PrimePalindrome.java
java PrimePalindrome
```

Output Screen

```
Enter a number: 131 is a prime palindrome
```

Result

The program demonstrates the listed syllabus concepts. The sample output is based on a compiled execution.